

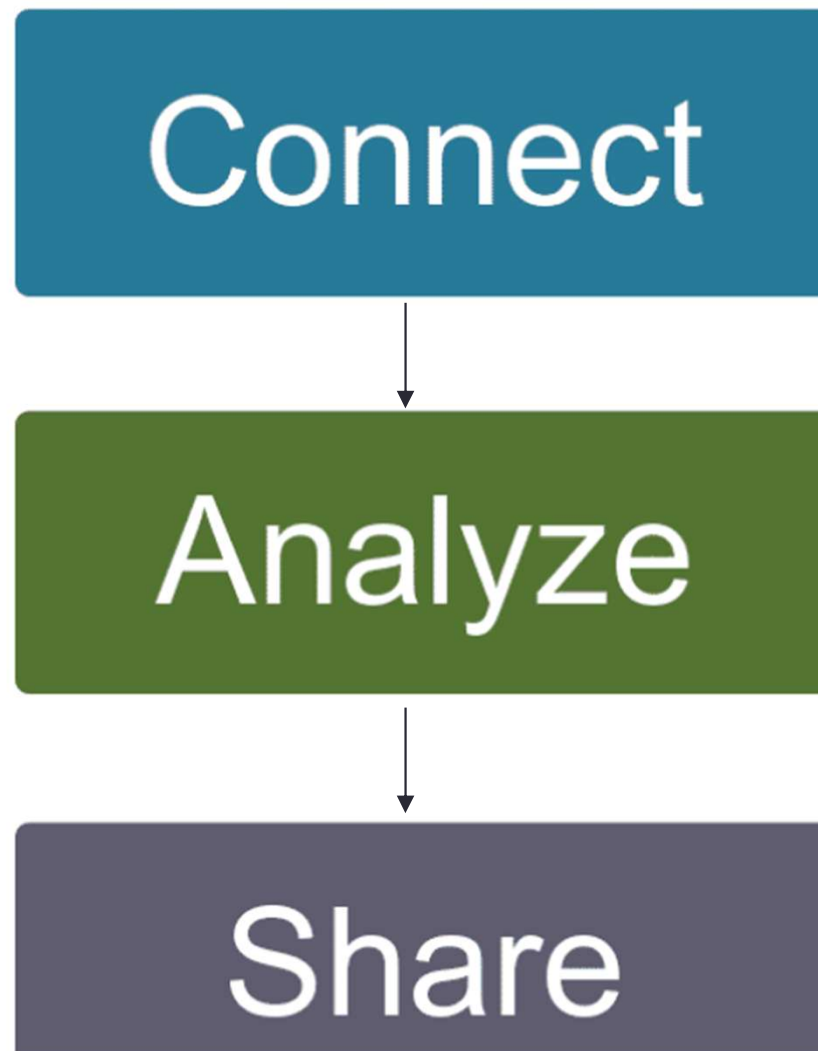
VISUALIZATION

Tableau Desktop – Connect

Installation

1. Download the latest version of Tableau Desktop:
<https://www.tableau.com/tft/activation>
2. Select “Download Tableau Desktop”
3. On the form, enter your school email address for Business E-mail and enter the name of your school for Organization.
4. Activate with your product key: **TCIZ-3615-B200-148D-4302**
5. Student Resource Page: <https://community.tableau.com/s/students>

Tableau Workflow



Outline

- Connect
- Analyze
- Share

Outline

- Connect
- Analyze
- Share

Data Sources

- Contains the data used to build visualizations.
 - Spreadsheet
 - Relational Databases
 - Cloud data
 - Other types of connections

Live vs Extract

■ Live

- Connecting live leaves the data in the database or source file.
- This is best when we want to leverage a high-performance database's capabilities, or to get up-to-the-second changes in data visualized in Tableau.
- However, sometimes connecting live can result in a slow experience, depending on the database.

■ Extract

- The other option is to extract the data into Tableau's high performance in-memory data engine.
- This can help when connecting to a slow database or to take query load off critical systems.
- We can also import only some of the data and bring in specific elements

Connecting to Data Sources

- After the connection, Tableau reads the data and assigns a data type to each field
- Users verify that Tableau has correctly assigned data types, and they can make changes on the **Data Source** page if the data type is wrong. They can also rename unclear or misspelled fields.
- Any changes made in Tableau are not written to the native data source. Changes are instead stored as metadata in a Tableau file called a Tableau Data Source, or .tds.
- After connecting to data and adjusting the metadata, users can open a sheet to begin building charts, analyzing data, and sharing results.

Data Source Page

The screenshot shows the Data Source Page interface with the following components:

- Connections:** A list of data sources, including 'Sample - Superstore' (Microsoft Excel).
- Sheets:** A list of worksheets in the selected workbook, including 'Orders', 'People', and 'Returns'.
- Metadata:** A section displaying the first 1,000 rows of data from the 'Orders' table. The data includes columns for Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country/Region, City, State, and Postal Code.
- Canvas:** A workspace for building data models, showing a diagram with a table named 'Orders' and a prompt to 'Need more data? Drag tables here to relate them: Learn more'.
- Data preview:** A table showing the first 1,000 rows of data from the 'Orders' table.

Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State	Postal Code
1	CA-2019-157108	11/8/2018	11/11/2018	Second Class	CS-67870	Chloe Guba	Consumer	United States	Portland	Oregon
2	CA-2019-157109	11/8/2018	11/11/2018	Second Class	CS-67870	Chloe Guba	Consumer	United States	Portland	Oregon
3	CA-2019-158088	6/15/2019	6/16/2019	Second Class	DS-53045	Dawn Van Hout	Corporate	United States	Los Angeles	California
4	US-2019-138998	10/11/2018	10/18/2018	Standard Class	SO-23320	Sean O'Sullivan	Consumer	United States	Fort Lauderdale	Florida
5	US-2019-138998	10/11/2018	10/18/2018	Standard Class	SO-23320	Sean O'Sullivan	Consumer	United States	Fort Lauderdale	Florida
6	CA-2017-115812	6/9/2017	6/14/2017	Standard Class	BM-11733	Brosina Hoffman	Consumer	United States	Los Angeles	California
7	CA-2017-115812	6/9/2017	6/14/2017	Standard Class	BM-11733	Brosina Hoffman	Consumer	United States	Los Angeles	California
8	CA-2017-115812	6/9/2017	6/14/2017	Standard Class	BM-11733	Brosina Hoffman	Consumer	United States	Los Angeles	California
9	CA-2017-115812	6/9/2017	6/14/2017	Standard Class	BM-11733	Brosina Hoffman	Consumer	United States	Los Angeles	California
10	CA-2017-115812	6/9/2017	6/14/2017	Standard Class	BM-11733	Brosina Hoffman	Consumer	United States	Los Angeles	California
11	CA-2017-115812	6/9/2017	6/14/2017	Standard Class	BM-11733	Brosina Hoffman	Consumer	United States	Los Angeles	California

Connections

The name of the data source

Sheets

The names of the worksheets in the workbook

Data preview

The first 1,000 rows of data

Metadata

Information about the data, such as the field name and data type

Canvas

Sheets or tables you are working with

Fields

- A field, also known as a column, is a single piece of information from a record in a data set.
- The qualitative fields that describe categories of data are in the top part of the pane, under **Dimensions**.
- The quantitative fields that measure categories of data are in the bottom part of the pane, under **Measures**.
- By default dimensions are discrete and measures are continuous.
- It's possible to have a numerical dimension be continuous, or to convert a measure to discrete (by right clicking and selecting “convert to discrete”).

Data Type

Icon	Data type
Abc	Text (string) values
	Date values
	Date & Time values
#	Numerical values
T F	Boolean values (relational only)
	Geographic values (used with maps)

Data can be organized into hierarchies

Hierarchy (Cross Table)

Category	Sub-Catego..	Manufacturer	Product Name	Profit	Sales
Furniture	Bookcases	Atlantic	Atlantic Metals ..	-311	545
			Atlantic Metals ..	780	7,540
			Atlantic Metals ..	-126	5,184
			Atlantic Metals ..	15	5,493
		Bestar	Bestar Classic B..	-613	1,898
		Bush	Bush Andora Bo..	136	3,318
			Bush Birmingha..	-118	825
			Bush Cubix Colle..	49	729
			Bush Heritage Pi..	-209	3,626
			Bush Mission Poi..	24	749
			Bush Saratoga C..	-13	120
			Bush Somerset C..	-56	1,264
			Bush Westfield C..	-97	649
			Bush Westfield C..	-20	1,427
		DMI	DMI Eclipse Exec..	90	11,047
		Global	Global Adaptabil..	207	3,448
		Hon	Hon 4-Shelf Met..	-192	1,474
			Hon Metal Bookc..	-27	1,570
			Hon Metal Bookc..	230	852
			Hon Metal Bookc..	363	1,320

Multiple Data Sources

Method	Description		Where created
Relationship 	Establishes the capability to use fields/columns from multiple tables.	Your data...  ...in a relationship: 	Logical layer on the canvas
Join 	Physically combines multiple tables by adding fields/columns.	Your data...  ...in a join: 	Physical layer on the canvas
Union 	Physically combines multiple tables by appending rows.	Your data...  ...in a union: 	Physical layer on the canvas
Blend 	Establishes linking field to use fields/columns from primary and aggregated secondary data sources.	Your data...  ...in a blend: 	Worksheet within a workbook

Relationships

Customer Table

Customer ID	Customer Name
1	Lisa Sanchez
2	Ahmed Abbas
3	Leena Gupta

Category Table

Category ID	Category Name
1	Containers
2	Spices
3	Drinkware
4	Utensils

Transaction Table

Transaction ID	Transaction Date	Customer ID
1	3/1/2021	1
2	3/8/2021	2
3	3/8/2021	1
4	3/15/2021	3

Transaction Details

Transaction ID	Product ID	Quantity Sold
1	1	5
1	4	2
2	6	1
2	6	2
2	2	11
3	7	1

Product Table

Product ID	Product Name	Unit Cost	Unit Price	Category ID
1	Small Spice Container	10.99	13.99	1
2	Medium Spice Container	14.99	17.99	1
3	Large Spice Container	19.99	23.99	1
4	Black Pepper	15.99	17.99	2
5	Cayenne Pepper	17.99	19.99	2
6	Sea Salt	10.99	12.99	2



Joins



The image shows a storefront for 'Floral Boutique' with a red and white striped awning. In the foreground, there are two tables. The left table is titled 'Price per dozen' and lists four flower types with their prices. The right table is titled 'Inventory' and lists five flower types with their inventory counts. The background features a pink floral wreath with the boutique's name.

Flower	Price
Rose	\$25
Lily	\$18
Carnation	\$10
Daisy	\$15

Flower	Inventory
Lily	50
Carnation	200
Daffodil	300
Tulip	140



Inner Join

Flowers that have prices and inventory.



Left Join

Flowers that have prices even if there is no inventory yet.



Right Join

All flowers with inventory even if they don't have prices.



Full Outer Join

All flowers even if they don't have prices or inventory.

Unions

- In Tableau, you can use a union to combine data that has a similar structure so that rows from one table are appended to the rows of another table.
- For example, you might want to combine suppliers in one table, who have last names starting with A-M, with the suppliers in another table, who have last names starting with N-Z, so you have a complete list of suppliers in one table.

2015

Date	Avg Temp	Rain	Max Wind
7/1/2015	78	0.6	
8/1/2015	79	0.5	13
9/1/2015	76	0.8	12
9/1/2016	75	1.0	11

2016

Date	Avg Temp	Rain	Max Wind
7/1/2016	76	0.8	13
8/1/2016	80	0.1	13
9/1/2016	75		11

2017

Date	Avg Temp	Rain	Max Wind
7/1/2017	79	0.5	12
8/1/2017		0.5	13
9/1/2017	78	1.0	11

Union Results

Date	Avg Temp	Rain	Max Wind
7/1/2015	78	0.6	<i>null</i>
8/1/2015	79	0.5	13
9/1/2015	76	0.8	12
7/1/2016	76	0.8	13
8/1/2016	80	0.1	13
9/1/2016	75	1.0	11
9/1/2016	75	<i>null</i>	11
7/1/2017	79	0.5	12
8/1/2017	<i>null</i>	0.5	13
9/1/2017	78	1.0	11

Blend

- Blends, unlike relationships or joins, never truly combine the data.
- Instead, blends query each data source independently, the results are aggregated to the appropriate level, then the results are presented visually together in the view.
- Because of this, blends can handle data sources that have different levels of detail, work with data sources that don't allow relationships or joins, and work with data sources that have already been published to Tableau Cloud or Tableau Server.
- Blend uses Left Join to combine the data sources.
- Use Cases
 - You would like the specific linking fields or the number of linking fields to vary on a sheet-by-sheet basis.
 - You want the primary data source to differ on a sheet-by-sheet basis.
 - You are using data sources, such as .json files and cube data, that cannot be combined with the default method of using a relationship.
 - You want to combine published data sources, but you do not have that capability, as in the case that the data steward has not made the structure of the data source available to edit.

Multiple Data Sources

Multiple Data Sources (Cross Table)

Region	Regional Manager	Profit	Sales
Central	Roxanne Rodriguez	39,865	503,171
East	Chuck Magee	94,883	691,828
South	Fred Suzuki	46,749	391,722
West	Sadie Pawthorne	110,799	739,814

Tableau Help

- <https://www.tableau.com/support/help>